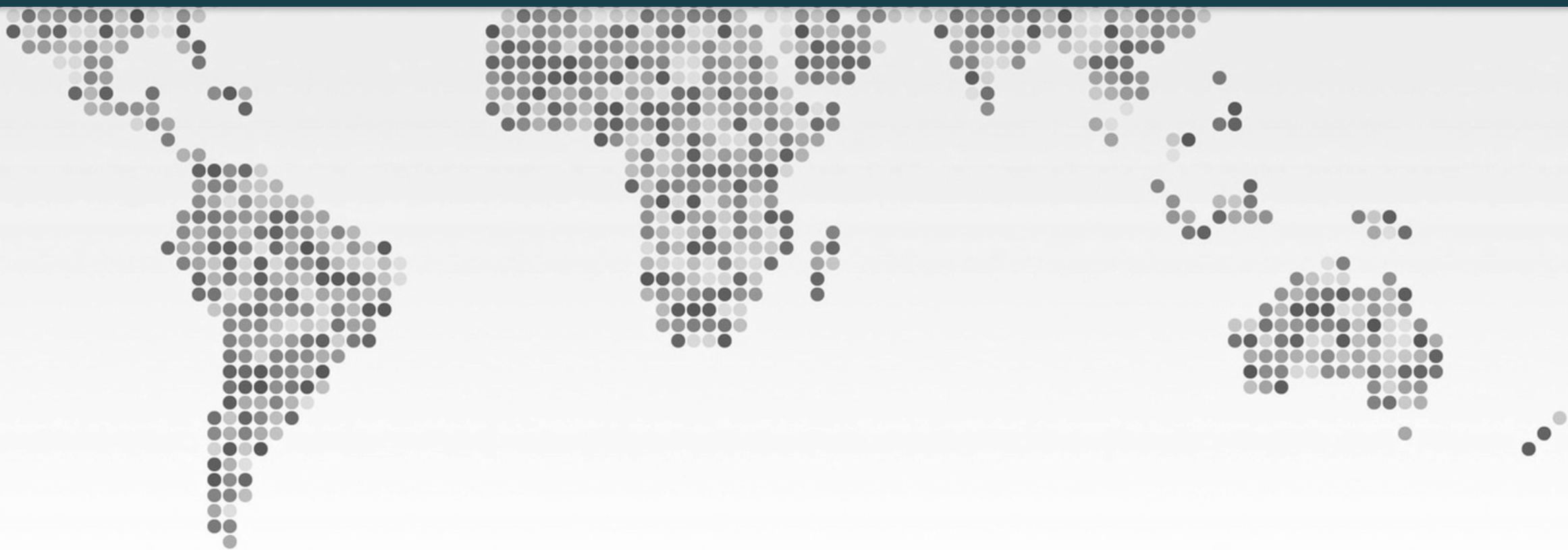


Software Engineering for Artificial Intelligence

Interpretability and Explainability of Artificial Intelligence Solutions



Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Why?

We have already mentioned the GDPR, the General Data Protection Regulation. Besides privacy-related regulations, the European Union introduced the *right* to an explanation of the decisions reached by automated mechanisms.

The interpretability of algorithms may help stakeholders in multiple manners:

- (1) *Achieve better outcomes*, e.g., Why was my message considered as spam? How can I improve my writing skills?
- (2) *Debugging*, e.g., What does the learner actually learn? What kind of data would make it better? Why did the system make a wrong prediction?
- (3) *Auditing*, e.g., inspect fairness properties, ensure predictions are based on objective criteria and/or reasonable rules.
- (4) *Curiosity, learning, discovery*, e.g., What drove our hiring decisions? What factors influence the demand of bike rentals?

Unfortunately, there is no formal definition of *interpretability*.

Interpretability. The degree to which a human can (1) understand the cause of a decision and (2) consistently predict the model's result.

Hence, we need to identify the properties enabling interpretability in order to identify metrics and methods to measure it. Also, we need to identify how humans would like to have explanations on the actions done by an artificial intelligence solution.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Why?

Let's start with defining the key terms of interpretability and explainability.

Interpretable models

Models that are intrinsically interpretable, i.e., models simple enough to understand (e.g., short decision trees, sparse linear models).

Explainability

It refers to *post-hoc* explanations, i.e., local or global explanations made after that a black-box model makes predictions.

Explanation

This is a justification of a single prediction, typically made in terms of natural language or charts.

Interpretability and explainability are often used inconsistently or interchangeably. Good explanations are *contrastive*. Even partial explanations are often effective in practice if they are posed in an opposite fashion with respect to predictions.

Why this, rather than a different prediction?

Counterfactuals are easier to understand, e.g.: “*Your loan application has been declined. If your savings account had had more than \$100 your loan application would be accepted.*”

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This is a *contrastive* explanation, as it explains what are the actions that the customer can do to have better predictions in the future

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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Good explanations are *social*. Different audiences might benefit from different explanations, hence an explainable approach should be *consistent* with prior belief of the explainee.

In addition, explanations are *selective*. Artificial intelligence methods, and machine learning in particular, rely on multiple features: as such, there are often long or multiple explanations for a prediction, thus requiring some form of selection.

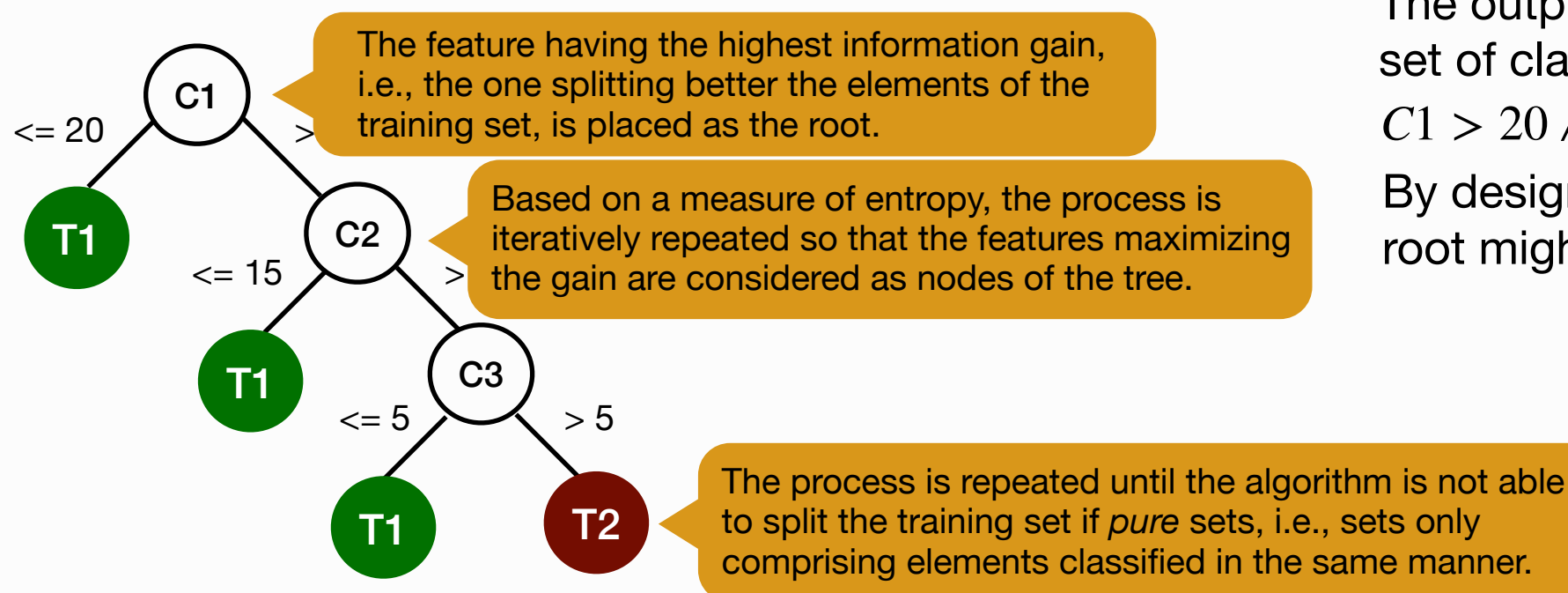
Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Inherently Interpretable Models

So, there is a key distinction to make between inherently interpretable models and post-hoc explanations.

The key example, that we already discussed in previous lectures, is the one of Decision Trees. Let's consider them again.

Engineering decisions for configuring machine learning algorithms



The output of a decision tree is a set of classification rules, e.g.:

$$C1 > 20 \wedge C2 > 15 \wedge C3 > 5 \Rightarrow T2$$

By design, a wrong selection of the root might lead to poor performance.

Decision Trees are easy to interpret up to a size: shorter trees are immediately interpretable by humans, who can also derive the importance of each feature.

Perhaps more importantly, it is possible to derive counterfactuals by considering the output rules. Yet, the instability of the model to small changes represent a problem.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

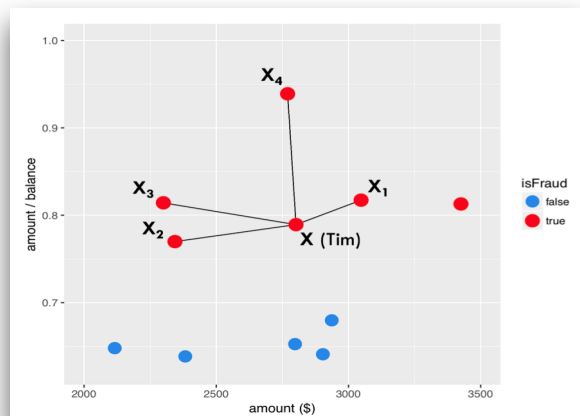
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Other examples of interpretable models are the k-nearest neighbor and association rules.

K-Nearest Neighbor



We already discussed this classifier when talking about SMOTE. This is an instance-based learning, that returns the most common class among the k nearest training data points.

It can be considered as inherently interpretable since one can interpret results by showing nearest neighbors. Of course, the interpretation assumes an understandable distance function and interpretable reference data points.

Example: *predict & explain car prices by showing similar sales.*

Association Rules

Starting from data presented in form of associations $X \rightarrow Y$, this is an unsupervised learning mechanism that extracts the so-called *frequent itemsets*, i.e., the most frequent associations in the data.

Example: $\{Diaper, Beer\} \rightarrow Milk$ (40% support, 66% confidence)

Percentage of associations (called transactions) in the dataset T that contain **both** items X and Y .

$$support(X \rightarrow Y) = Prob(A \cup B)$$

Software Engineering for Artificial Intelligence: Interpretability and Explainability

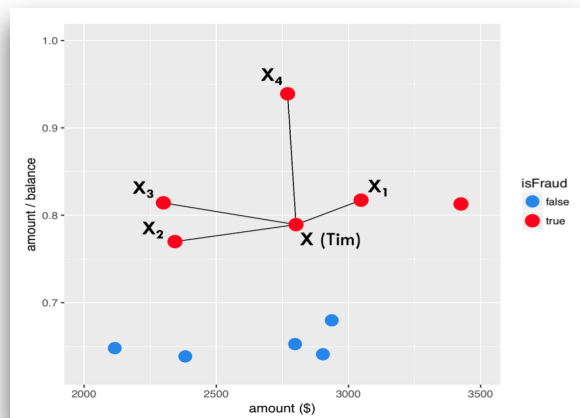
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Percentage of transactions in T , containing X , that **also** contain Y .

$$confidence(X \rightarrow Y) = Prob(B | A)$$

Software Engineering for Artificial Intelligence: Interpretability and Explainability

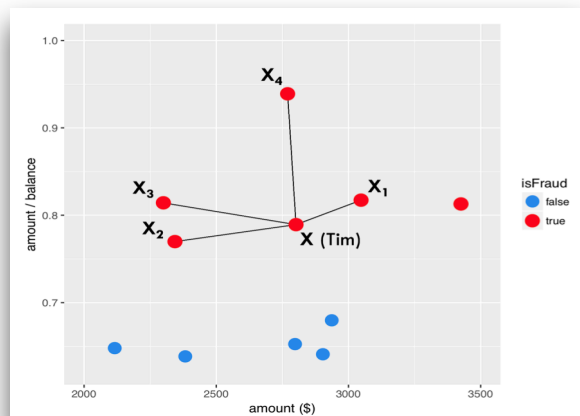
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Example: $\{Diaper, Beer\} \rightarrow Milk$ (40% support, 66% confidence)

If there are few rules and these are simple enough, then the approach can be considered inherently interpretable. More in general, all the approaches elaborating data to extract *if-then* rules belong to this category.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Inherently Interpretable Models

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Other examples of interpretable models are the k-nearest neighbor and association rules.

Research in interpretable models is attempting to approach the problem under various perspectives. For instance, by learning *sparse constrained models*: these are methods for performing high-dimensional non-parametric regression, i.e., these methods attempt to reduce the feature space to limit the scope of artificial intelligence models to the essential.

As you can imagine, *feature engineering* plays a crucial role here.

Feature Selection. Process through which the most relevant features are identified, while the non-relevant or redundant are filtered out.

If we are able to select the most relevant features, then the interpretability of the models naturally increases.

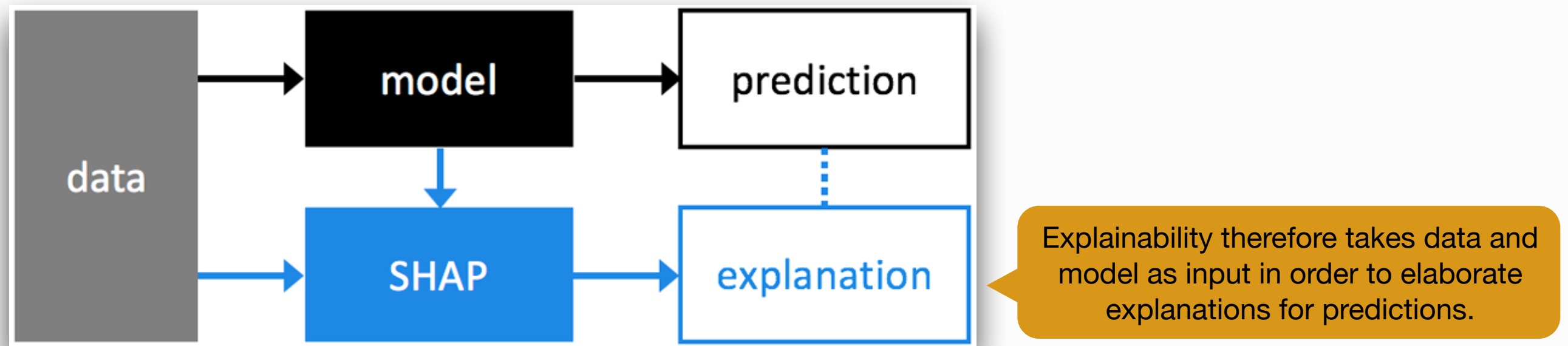
Recent advances attempted to include the concept of *domain-specificity* to feature engineering, namely the exploitation of contextual information to improve the agnostic feature selection methods.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

Nonetheless, making artificial intelligence models inherently interpretable is not always possible, e.g., consider the case of deep neural network.

That is why we cannot limit ourselves to working with features and experimenting with simpler models, but require techniques that can work on top of the model outputs.



The two most common approaches relate to the use of *global* and *local surrogates*.

Global surrogate. This is an interpretable model that is trained to *approximate* the predictions of a black box model.

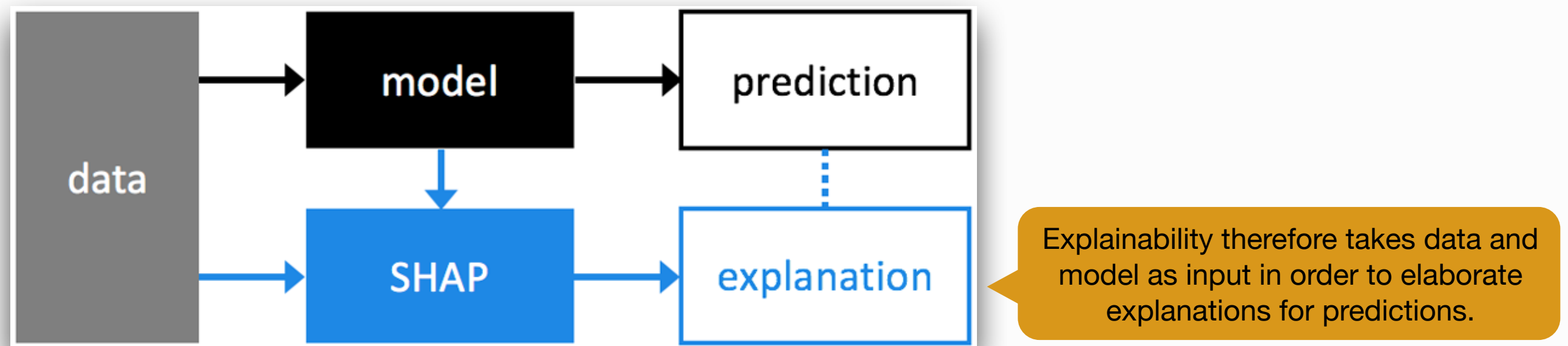
Local surrogate. This is an interpretable model to explain *individual predictions* of black box machine learning models.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

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So, global models aim at approximating the model to explain, while local models aim at finding explanations for individual predictions.

In any case, the important thing to notice is that both treat models as *black-boxes*! Surrogates therefore attempt to observe models by querying them, without having access to internal information.

Depending on the query approach, we distinguish global versus local models.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

Let's take a closer look at these surrogates, starting from the global ones.

With global surrogates we can draw conclusions about the black box model by interpreting the surrogate model. In other terms, we are solving machine learning interpretability by using more machine learning.

The idea is to create a cheap and fast surrogate model of the outcome to estimate the rationale behind an outcome which is expensive, time-consuming or difficult to measure.

We want to *approximate* our black box prediction function f as closely as possible with the surrogate model prediction function g , under the constraint that g is interpretable.

For the function g , any *interpretable* model can be used, e.g., a linear regression or a decision tree. Hence, we approximate a complex model with a simpler one.

Training a surrogate model is a *model-agnostic* method, since it does not require any information about the inner workings of the black-box model, only access to data and the prediction function is necessary. That means that, if you change the black-box model, you can still reuse the surrogate —> this is the reason why you can use available explainable AI technique on any machine learner.

Before looking at how to build a surrogate model, there is another key observation to make: the model performance of the underlying black-box model, i.e., its accuracy, does not play a role in training the surrogate model. The interpretation of the surrogate model is still valid but, of course, interpretation becomes irrelevant if the black-box model is bad.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

The general procedure to create global surrogates is the following:

Select a dataset X

The first step consists of using the same dataset used for training the black box model or a new dataset from the same distribution. You could even select a subset of the data or a grid of points, depending on your application.

This is already quite important! For complex tasks, your dataset might be hard to treat (e.g., images). Hence, the ability to select a suitable subset impacts the ability to explain the predictions of the black-box model.

Get predictions

For the selected dataset X, get the predictions of the black box model.

Interpretability step

Select an interpretable model type (linear model, decision tree, etc.).

Training step

Train the interpretable model on the dataset X and its predictions.

Done. You have a surrogate model. Nonetheless, you might want to measure how well the surrogate model replicates the predictions of the black-box model. The measure typically used is the R^2 , i.e., percentage of variance that is captured by the surrogate model. If R^2 is close to 1, then the surrogate approximates the model very well.

Someone might ask: *Why should I have a complex black-box model if a simpler surrogate is able to replicate it well?*

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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Someone might ask: *Why should I have a surrogate model if the black box model is able to replicate it well?*

1. Do not forget that surrogate models can be built starting from a subset of the training data!
2. There might be other reasons to use complex models (e.g., non-functional requirements).

But, in any case, if R^2 is close to 1, replacing the complex model with the surrogate is definitively an option!

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

So, to sum up: with global surrogates we create a simpler model to explain a more complex model. Nothing more, nothing less.

Advantages

The surrogate model is *flexible*, as any machine learning model can be explained by means of global models. At the same time, you can change the surrogate models as you like based on the fitness that it has with the model to explain.

The global modeling is *intuitive* and *straightforward*. On the one hand, it is easy to implement, as it requires basic knowledge of machine learning. On the other hand, it is also easy to explain to people not familiar with data science or machine learning.

Disadvantages

You must be aware that you draw *conclusions about the model and not about the data*, since the surrogate model never sees the real outcome.

It is not clear what the best cut-off for R^2 is in order to be confident that the surrogate model is close enough to the black box model.

It could happen that the interpretable model is very *close for one subset of the dataset, but widely divergent for another subset*. In this case the interpretation for the simple model would not be equally good for all data points.

Some people argue that there are, in general, *no intrinsically interpretable models* (including even linear models and decision trees) and that it would even be dangerous to have an illusion of interpretability. No one knows, it is still open to interpretation!

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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More information about global models are available in the additional material. The most widely known implementation is SHAP (SHapley Additive exPlanations): <https://shap.readthedocs.io/en/latest/generated/shap.Explainer.html>

A Unified Approach to Interpreting Model Predictions

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Abstract

Understanding why a model makes a certain prediction can be as crucial as the prediction's accuracy in many applications. However, the highest accuracy for large modern datasets is often achieved by complex models that even experts struggle to interpret, such as ensemble or deep learning models, creating a tension between *accuracy* and *interpretability*. In response, various methods have recently been proposed to help users interpret the predictions of complex models, but it is often unclear how these methods are related and when one method is preferable over another. To address this problem, we present a unified framework for interpreting predictions, SHAP (SHapley Additive exPlanations). SHAP assigns each feature an importance value for a particular prediction. Its novel components include: (1) the identification of a new class of additive feature importance measures, and (2) theoretical results showing there is a unique solution in this class with a set of desirable properties. The *new* class unifies six existing methods, notable because several recent methods in the class lack the proposed desirable properties. Based on insights from this unification, we present new methods that show improved computational performance and/or better consistency with human intuition than previous approaches.

1 Introduction

The ability to correctly interpret a prediction model's output is extremely important. It engenders appropriate user trust, provides insight into how a model may be improved, and supports understanding of the process being modeled. In some applications, simple models (e.g., linear models) are often preferred for their ease of interpretation, even if they may be less accurate than complex ones. However, the growing availability of big data has increased the benefits of using complex models, so bringing to the forefront the trade-off between accuracy and interpretability of a model's output. A wide variety of different methods have been recently proposed to address this issue [5, 8, 9, 3, 4, 1]. But an understanding of how these methods relate and when one method is preferable to another is still lacking.

Here, we present a novel unified approach to interpreting model predictions.¹ Our approach leads to interesting results that bring clarity to the growing space of methods:

...pective of viewing *any* explanation of a model's prediction as a model itself, *explanation model*. This lets us define the class of *additive feature attribution* which unifies six current methods.

<https://github.com/slundberg/shap>

Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

Let's now consider the case of local models to explainability.

With local interpretable surrogates we create models that are used to explain individual predictions of black-box machine learning models.

The seminal paper was the one presenting the concept of local interpretable model-agnostic explanations (LIME), which is also a well-known Python library: <https://github.com/marcotcr/lime>.

First, forget about the training data and imagine you only have the black-box model where you can input data points and get the predictions of the model. You can probe the box as often as you want.

Your goal is to understand why the machine learning model made a certain prediction. LIME tests what happens to the predictions when you give *variations* of your data into the machine learning model.

As such, LIME generates a new dataset consisting of perturbed samples and the corresponding predictions of the black box model.

On this new dataset LIME then trains an interpretable model, which is weighted by the proximity of the sampled instances to the instance of interest.

The learned model should be a good approximation of the machine learning model predictions locally, but it does not have to be a good global approximation. This kind of accuracy is also called local fidelity.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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Your goal is to understand why the model predicts a certain value. LIME tests what happens to the predictions of the machine learning model.

As such, LIME generates a new dataset of data points and their corresponding predictions of the black-box model.

Because of this property, you try to explain a model by learning from *local* variations of data, rather than from *global* training data. You test the model with perturbations, so that you can learn the way the model acts. Local learning can be seen as a way to interpret the black-box model by multiple interactions with it - similar to what humans do: we learn other people by observing how they react to various stimuli.

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Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

The general procedure to create local surrogates is the following:

Select an instance X

The first step consists of selecting an instance or more instances of interest for which you want to have an explanation of its black box prediction.

Perturbation step

Perturb your dataset and get the black box predictions for these new points.

Training step

Train an interpretable model weighting the new points based on their proximity to the instance of interest, i.e., you weight the model based on how close the predictions of these samples are with respect to the black-box predictions.

In other terms, you train a surrogate model selecting a subset of instances from the black-box model, varying them and understand what are the weights that led the black-box model predict in certain manners.

The surrogate model therefore aims at learning local weights. Its accuracy depends, once again, on how data get variated.

This depends on the type of data. For text and images, the solution is to turn single words or pixels on or off. In the case of tabular data, LIME creates new samples by perturbing each feature individually, drawing from a normal distribution with mean and standard deviation taken from the feature.

Existing instruments allow to parameterize the number of instances to interpret, other than the number of features to use for the explanation.

Linear regression is typically the most widely used interpretable model used.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

So, to sum up: with local surrogates we create a simpler model to explain the predictions made on specific instances of the more complex model.

Advantages

LIME is one of the few methods that works for tabular data, text and images.

The fidelity measure, which indicates how well the interpretable model approximates the black-box predictions, gives us a good idea of how reliable the interpretable model is in explaining the black-box predictions in the neighborhood of the data instance of interest.

Disadvantages

LIME explanations can be manipulated by the data scientist to hide biases, i.e., various malicious attacks can be done to bias the explanations provided. The possibility of manipulation makes it more difficult to trust explanations generated with LIME.

The instability of the explanations. The explanations of two very close points vary greatly because of the perturbation step, other than depending on the surrogate model selected.

Local surrogate models, with LIME as a concrete implementation, are very promising. But the method is still in development phase and many problems need to be solved before it can be safely applied.

So, up to now, global models should be preferred to local ones. Nonetheless, our argumentation should make clear that good explainability is still far from being obtained and quite hard to obtain anyway.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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“Why Should I Trust You?” Explaining the Predictions of Any Classifier

Marco Tulio Ribeiro
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ABSTRACT

Despite widespread adoption, machine learning models remain mostly black boxes. Understanding the reasons behind predictions is, however, quite important in assessing *trust*, which is fundamental if one plans to take action based on a prediction, or when choosing whether to deploy a new model. Such understanding also provides insights into the model, which can be used to transform an untrustworthy model or prediction into a trustworthy one.

In this work, we propose LIME, a novel explanation technique that explains the predictions of *any* classifier in an interpretable and faithful manner, by learning an interpretable model locally around the prediction. We also propose a method to explain models by presenting representative individual predictions and their explanations in a non-redundant way, framing the task as a submodular optimization problem. We demonstrate the flexibility of these methods by explaining different models for text (e.g. random forests) and image classification (e.g. neural networks). We show the utility of explanations via novel experiments, both simulated and with human subjects, on various scenarios that require trust: deciding if one should trust a prediction, choosing between models, improving an untrustworthy classifier, and identifying why a classifier should not be trusted.

1. INTRODUCTION

Machine learning is at the core of many recent advances in science and technology. Unfortunately, the important role of humans is an oft-overlooked aspect in the field. Whether humans are directly using machine learning classifiers as tools, or are deploying models within other products, a vital concern remains: *if the users do not trust a model or a prediction, they will not use it*. It is important to differentiate between two different (but related) definitions of trust: (1) *trusting a prediction*, i.e. whether a user trusts an individual prediction sufficiently to take some action based on it, and (2) *trusting a model*, i.e. whether the user trusts a model to behave in reasonable ways if deployed. Both are directly impacted by

how much the human understands a model's behaviour, as opposed to seeing it as a black box.

Determining trust in individual predictions is an important problem when the model is used for decision making. When using machine learning for medical diagnosis [6] or terrorism detection, for example, predictions cannot be acted upon on blind faith, as the consequences may be catastrophic.

Apart from trusting individual predictions, there is also a need to evaluate the model as a whole before deploying it “in the wild”. To make this decision, users need to be confident that the model will perform well on real-world data, according to the metrics of interest. Currently, models are evaluated using accuracy metrics on an available validation dataset. However, real-world data is often significantly different, and further, the evaluation metric may not be indicative of the product's goal. Inspecting individual predictions and their explanations is a worthwhile solution, in addition to such metrics. In this case, it is important to aid users by suggesting which instances to inspect, especially for large datasets.

In this paper, we propose providing explanations for individual predictions as a solution to the “trusting a prediction” problem, and selecting multiple such predictions (and explanations) as a solution to the “trusting the model” problem. Our main contributions are summarized as follows.

- LIME, an algorithm that can explain the predictions of *any* classifier or regressor in a faithful way, by approximating it locally with an interpretable model.
- SP-LIME, a method that selects a set of representative instances with explanations to address the “trusting the model” problem, via submodular optimization.
- Comprehensive evaluation with simulated and human subjects, where we measure the impact of explanations on trust and associated tasks. In our experiments, non-experts using LIME are able to pick which classifier from a pair generalizes better in the real world. Further, they are able to greatly improve an untrustworthy classifier trained on 20 newsgroups, by doing feature engineering using LIME. We also show how understanding the predictions of a neural network on images helps practitioners know when and why they should not trust a model.

2. THE CASE FOR EXPLANATIONS

By “explaining a prediction”, we mean presenting textual or visual artifacts that provide qualitative understanding of the relationship between the instance's components (e.g. words in text, patches in an image) and the model's prediction. We argue that explaining predictions is an important aspect in

More information about local models are available in the additional material. LIME (Local Interpretable Model-agnostic Explanations) is also a well-known Python library: <https://github.com/marcotcr/lime>.

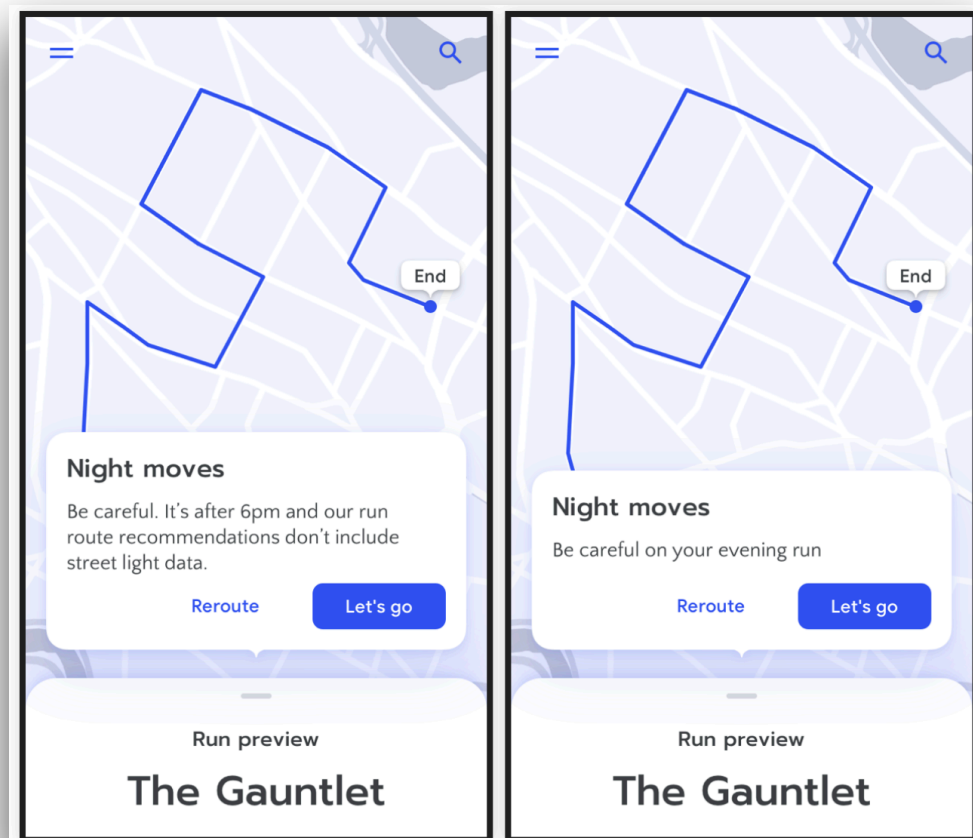
Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Interaction Design

Independently from the method you use to create interpretable artificial intelligence models, it is worth reporting that interaction design plays a role in the way we present the information coming from explainable models.

As you understood, explainable models use interpretable models on subsets of data or specific instances. Yet, they are still models that need to be presented in a way that enables actionable decisions.

Natural language is typically used to translate interpretable rules into an actionable text. We do not have time to elaborate on NLP, but some concepts from human-computer interaction might be relevant.



Tell the user when a lack of data might mean they will need to use their own judgment. Do not be afraid to admit when a lack of data could affect the quality of the AI recommendations.

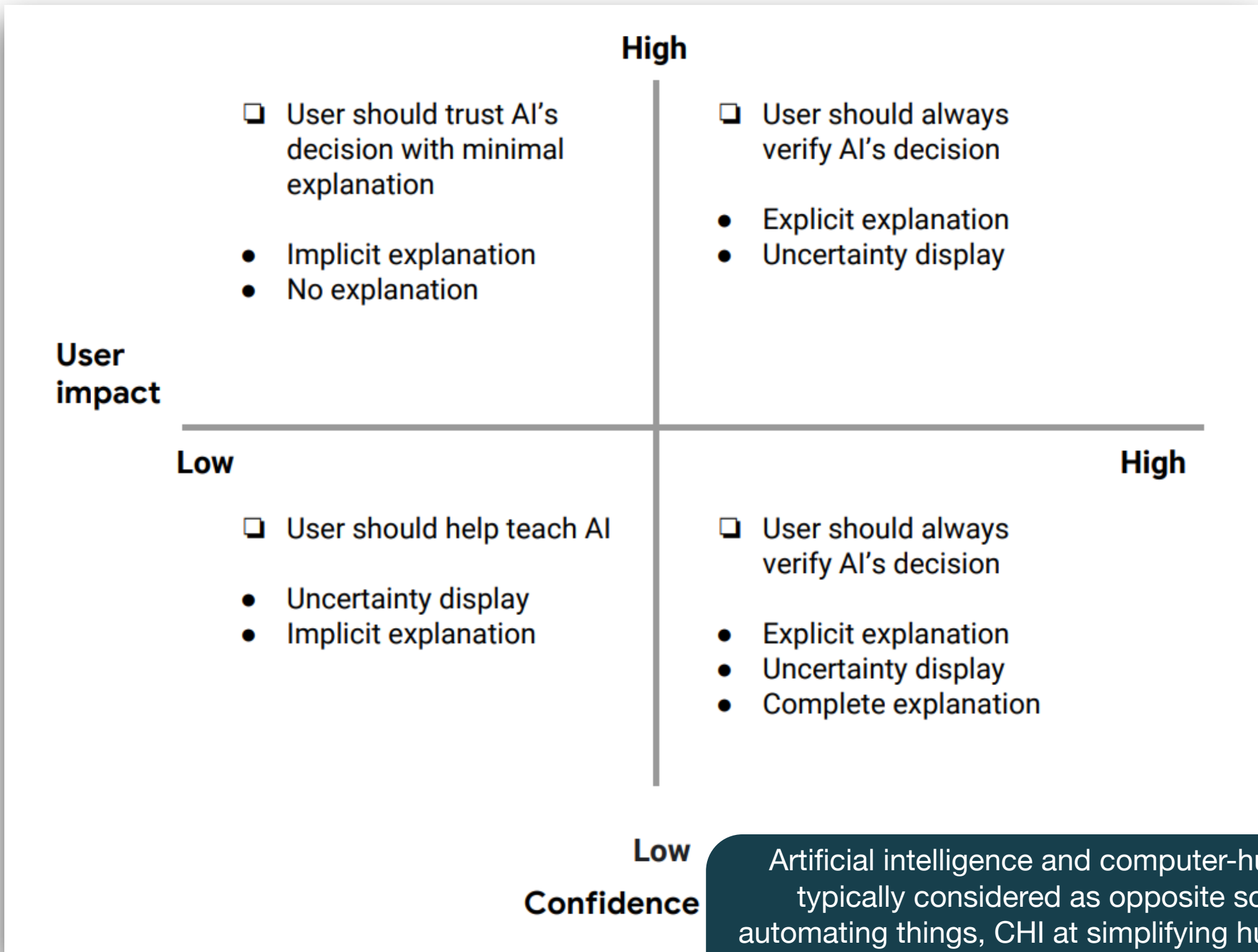
Give the user details about why a prediction was made in a high stakes scenario.

In other terms, use natural language processing by keeping interaction design principles into account, implementing mechanisms for actionable AI.

Interaction design indeed increases the confidence of users with respect to AI!

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Interaction Design



Artificial intelligence and computer-human interaction are typically considered as opposite sciences. AI aims at automating things, CHI at simplifying human actions. Yet, CHI can and should be combined with AI to make AI actionable!

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: What about GenAI?

As a “minor” note, it is worth remarking that explainability does not stop to LIME and SHAP. Explainability is an even harder concern when it comes to images (saliency, Grad-CAM, or LIME are methods to enact explainability) and GenAI systems.

Especially for GenAI systems, there are some additional points to remark:

Free-form outputs

The model generates **unstructured text**, not fixed labels, making explanations harder to standardize.

Stochastic behavior

The same input can produce **different outputs** due to randomness in generation.

Context-dependence

Outputs depend heavily on **prompt wording** and **conversation history**, not just the input itself.

No clear features

The model does not learn a clear boundary between classes, but rather **probabilistic token generation**.

In essence, the problem shifts from “*Why this prediction?*” to “*Why this generation?*”
Needless to say that the problem is an open one, with only a few approaches available...
How would you solve it?



In the following, some preliminary solutions; however, each of them brings key assumptions and limitations

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: What about GenAI?

Prompt-based explanations

Prompt-based explanations are explanations obtained by **explicitly asking a GenAI model** to describe its reasoning process in natural language.

The key idea is that the model may **self-generate a rationale** based on its internal representations and training. *What is the assumption here?* The explanation may sound **plausible**, but not always **faithful** (do not reflect the true internal reasoning). In other terms, the explanation may be hallucinated!

Plausibility ≠ Faithfulness

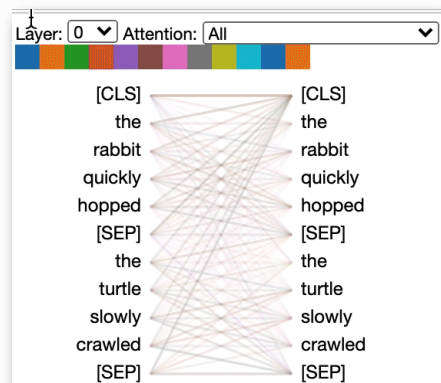
How to Check If an Explanation Is Faithful? Consistency Under Perturbation, e.g., input testing? Example-based, e.g., retrieve similar training examples?

Chain-of-Thoughts

Do you remember CoT? Well, in some sense it may be considered as an explainability method, as the model provides its own reasoning to generation.

Attention Visualization

AT is a technique to highlight which input tokens the model focuses on when generating an output, by inspecting attention weights.



Attention ≠ Explanation

How to Check If an Explanation Is Faithful?
Again, perturbation? Examples?

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: What about GenAI?

Or perhaps something more...



Software Engineering for Artificial Intelligence: Interpretability and Explainability

Before leaving... What Should You Remember?

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Why?

We have already mentioned the GDPR, the General Data Protection Regulation. Besides privacy-related regulations, the European Union introduced the *right* to an explanation of the decisions reached by automated mechanisms.

The interpretability of algorithms may help stakeholders in multiple manners:

- (1) *Achieve better outcomes*, e.g., Why was my message considered as spam? How can I improve my writing skills?
- (2) *Debugging*, e.g., What does the learner actually learn? What kind of data would make it better? Why did the system make a wrong prediction?
- (3) *Auditing*, e.g., inspect fairness properties, ensure predictions are based on objective criteria and/or reasonable rules.
- (4) *Curiosity, learning, discovery*, e.g., What drove our hiring decisions? What factors influence the demand of bike rentals?

Unfortunately, there is no formal definition of *interpretability*.

Interpretability. The degree to which a human can (1) understand the cause of a decision and (2) consistently predict the model's result.

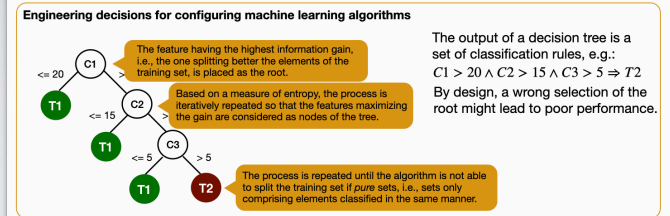
Hence, we need to identify the properties enabling interpretability in order to identify metrics and methods to measure it. Also, we need to identify how humans would like to have explanations on the actions done by an artificial intelligence solution.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Inherently Interpretable Models

So, there is a key distinction to make between inherently interpretable models and post-hoc explanations.

The key example, that we already discussed in previous lectures, is the one of Decision Trees. Let's consider them again.



Decision Trees are easy to interpret up to a size: shorter trees are immediately interpretable by humans, who can also derive the importance of each feature. Perhaps more importantly, it is possible to derive counterfactuals by considering the output rules. Yet, the instability of the model to small changes represent a problem.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

So, to sum up: with local surrogates we create a simpler model to explain the predictions made on specific instances of the more complex model.

Advantages
LIME is one of the few methods that works for tabular data, text and images. The fidelity measure, which indicates how well the interpretable model approximates the black-box predictions, gives us a good idea of how reliable the interpretable model is in explaining the black-box predictions in the neighborhood of the data instance of interest.

Disadvantages
LIME explanations can be manipulated by the data scientist to hide biases, i.e., various malicious attacks can be done to bias the explanations provided. The possibility of manipulation makes it more difficult to trust explanations generated with LIME. The instability of the explanations. The explanations of two very close points vary greatly because of the perturbation step, other than depending on the surrogate model selected.

Local surrogate models, with LIME as a concrete implementation, are very promising. But the method is still in development phase and many problems need to be solved before it can be safely applied. So, up to now, global models should be preferred to local ones. Nonetheless, our argumentation should make clear that good explainability is still far from being obtained and quite hard to obtain anyway.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Why?

Let's start with defining the key terms of interpretability and explainability.

Interpretable models	Models that are intrinsically interpretable, i.e., models simple enough to understand (e.g., short decision trees, sparse linear models).
Explainability	It refers to <i>post-hoc</i> explanations, i.e., local or global explanations made after that a black-box model makes predictions.
Explanation	This is a justification of a single prediction, typically made in terms of natural language or charts.

Interpretability and explainability are often used inconsistently or interchangeably. Good explanations are *contrastive*. Even partial explanations are often effective in practice if they are posed in an opposite fashion with respect to predictions.

Why this, rather than a different prediction?
Counterfactuals are easier to understand, e.g.: "Your loan application has been declined. If your savings account had had more than \$100 your loan application would be accepted."

Good explanations are *social*. Different audiences might benefit from different explanations, hence an explainable approach should be *consistent* with prior belief of the explainee.

In addition, explanations are *selective*. Artificial intelligence methods, and machine learning in particular, rely on multiple features: as such, there are often long or multiple explanations for a prediction, thus requiring some form of selection.

Software Engineering for Artificial Intelligence: Interpretability and Explainability

Interpretability and Explainability of Artificial Intelligence Solutions: Post-Hoc Explanations

So, to sum up: with global surrogates we create a simpler model to explain a more complex model. Nothing more, nothing less.

Advantages
The surrogate model is *flexible*, as any machine learning model can be explained by means of global models. At the same time, you can change the surrogate models as you like based on the fitness that it has with the model to explain. The global modeling is *intuitive* and *straightforward*. On the one hand, it is easy to implement, as it requires basic knowledge of machine learning. On the other hand, it is also easy to explain to people not familiar with data science or machine learning.

Disadvantages
You must be aware that you draw *conclusions about the model and not about the data*, since the surrogate model never sees the real outcome. It is not clear what the best cut-off for R^2 is in order to be confident that the surrogate model is close enough to the black box model. It could happen that the interpretable model is very *close for one subset of the dataset, but widely divergent for another subset*. In this case the interpretation for the simple model would not be equally good for all data points.

Some people argue that there are, in general, *no intrinsically interpretable models* (including even linear models and decision trees) and that it would even be dangerous to have an illusion of interpretability. No one knows, it is still open to interpretation!

Software Engineering for Artificial Intelligence: Interpretability and Explainability

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Software Engineering for Artificial Intelligence

Interpretability and Explainability of Artificial Intelligence Solutions

